# AQA 3.2.4 Properties of Period 3 elements and their oxides

*AQA A-level Chemistry · Revision questions · Sheet 5 of 6*

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

## Sheet 5 of 6

*18 marks · about 25 minutes*

**Key information**

- Basic oxides react with acids to give salt + water; acidic oxides react with bases (e.g. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$)
- $Na_2O$ and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)
- $Al_2O_3$ is insoluble and amphoteric; $SiO_2$ is insoluble, giant covalent and acidic (reacts with hot conc NaOH)
- $P_4O_{10}$, $SO_2$, $SO_3$ are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$
- Melting points: giant ionic and giant covalent oxides are high; simple molecular oxides are low (weak van der Waals forces)

### Section A: Short-answer questions

**01** Solutions of sulfur dioxide and sulfur trioxide with the same concentration are prepared. Suggest why the solution of sulfur dioxide has a higher pH.

**[1 mark]**

**02** Suggest why aluminium oxide and silicon dioxide do not change the pH of water.

**[1 mark]**

**03** Draw the displayed formula of the undissociated acid formed when sulfur dioxide reacts with water.

**[1 mark]**

**04** Which statement is correct?

**[1 mark]**

**A** A minimum of 0.5 mol of water is needed to react completely with 1 mol of $Na_2O$

**B** A solution of hydrochloric acid containing 1 mol of HCl reacts with 1 mol of MgO to form a solution with pH 7

**C** A solution of hydrochloric acid containing 6 mol of HCl reacts completely with 1 mol of $Al_2O_3$

**D** A minimum of 4 mol of water is needed to react completely with 1 mol of $P_4O_{10}$

**05** Give an equation for the reaction of silicon with oxygen.

**[1 mark]**

**06** Which Period 3 oxide has a giant covalent structure?

**[1 mark]**

**A** MgO

**B** $Al_2O_3$

**C** $SiO_2$

**D** $SO_2$

### Section B: Longer questions

**07** This question is about the oxides of the Period 3 elements sodium to sulfur.

**07.1** Give the formula of the oxide that has a giant ionic structure and forms the most alkaline solution in water. Give an equation for its reaction with water.

**[2 marks]**

**07.2** Aluminium oxide is amphoteric. Give an equation for its reaction with hydrochloric acid and an equation for its reaction with sodium hydroxide solution.

**[2 marks]**

**07.3** Explain why silicon dioxide has a very high melting point.

**[2 marks]**

**07.4** Give an equation for the reaction of sulfur trioxide with water and suggest a pH for the solution formed.

**[2 marks]**

**08** This question is about elements in Period 3 and their compounds.

**08.1** When a piece of sodium is added to 200 $cm^3$ of water in a large beaker a vigorous reaction occurs. The temperature of the water increases by 25 °C. Give an equation, including state symbols, for the reaction of sodium with water. Suggest why it is dangerous to react a similar piece of sodium with 10 $cm^3$ of water in a boiling tube.

**[2 marks]**

**08.2** Give an equation for the reaction of phosphorus(V) oxide with water. Suggest a pH for the solution formed.

**[2 marks]**

# Mark Scheme

*M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.*

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| Question | Marking guidance | Comments | Mark |
|---|---|---|---|
| 01 | $H_2SO_3$ is a weak acid / only partially dissociates, whereas $H_2SO_4$ is a strong acid | *Source: Practice question (written for this set)* | 1 |
| 02 | They are insoluble in water | *Source: Practice question (written for this set)* | 1 |
| 03 | Displayed formula of $H_2SO_3$: S bonded to one =O and two –O–H groups (all bonds shown) | *Source: 7405/1 Jun 2017 Q06.7*<br>Ignore lone pairs. Answer written from the MS diagram | 1 |
| 04 | **C** | *Source: 7405/3 Jun 2025 Q26* | 1 |
| 05 | $Si + O_2 \rightarrow SiO_2$ | *Source: Practice question (written for this set)* | 1 |
| 06 | **C** | *Source: Practice question (written for this set)* | 1 |
| 07.1 | $Na_2O$<br>$Na_2O + H_2O \rightarrow 2NaOH$ | *Source: Practice question (written for this set)* | 2 |
| 07.2 | $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$<br>$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$ | Allow ionic equations | 2 |
| 07.3 | M1 giant covalent / macromolecular<br>M2 many strong covalent bonds must be broken, requiring a lot of energy | | 2 |
| 07.4 | $SO_3 + H_2O \rightarrow H_2SO_4$<br>pH 0–1 | | 2 |
| 08.1 | $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$<br>Temperature will go up more / reactants could shoot out of the tube / could explode / hydrogen could ignite | *Source: 7405/1 Nov 2021 Q03.1–03.2*<br>Ignore reaction is exothermic / vigorous | 2 |
| 08.2 | $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$<br>pH –1 to +1 | Do not allow equations with $P_2O_5$ | 2 |